

DEBRE BERHAN UNIVERSITY College Computing Department of Information Technology Course Syllabus	
Program	BSc in Information Technology
Course Name	Internet Programming I
Course Code	ITec2092
Year/Semester	2 year/ semester II
Credit hours (ECTS)	5
Lecturing Hours	30 hrs.
Laboratory Hours	45 hrs.
Assessment hours	15 hrs.
Home study	45 hrs.
Total Notional Hours	135 hrs.
Pre-requisite	Computer programming
Academic Staff Name	Eskedar Y.
Course Summary	The course is a comprehensive that teaches students the essential foundations of web development. Students will learn how to create static web pages, interact with API, and build web applications. The course covers HTML, CSS and JavaScript, and includes an overview of the internet, the World Wide Web, and hypertext. By the end of the course, students will have a strong understanding of these key web development technologies and be equipped to create static and interactive websites. Emphasis is placed on practical application and teamwork.

I. Course Learning Outcome.

At the end of the course, students will be able to:

CLO1: Describe and demonstrate Web, standards, protocols, technologies, architectures and emerging ideas.

CLO2: Differentiate and apply frontend technologies such as HTML, CSS and JS according to UI/UX theory

CLO3: Design and develop well-structured, easily maintained, standards-compliant, interactive and responsive web application

CLO4: Debug, test and troubleshoot problems and validate HTML code

CLO5: Apply advanced programming concepts and explore frameworks and libraries for efficient web development.

CLO6: Work effectively in a team environment and use collaboration tools.

II. Mapping CLO and PLO

	PEO1	PEO2	PEO3	PEO4	PEO5	PLO6	Assessment method
CLO1	X						Demonstration, Mid-Exam, Final Exam
CLO2			X				Project, Mid-Exam, Final Exam
CLO3		X					Project, Final Exam
CLO4				X			Project, Final Exam
CLO5				X			Project, Final Exam
CLO6						X	Project

III. Course Content, CLO and teaching methods mapping

Course Content	CLO	Intervention methods
Chapter 1: Overview of the internet and www (8 hrs.) 1.1.Internet History, uses, and services. 1.2.Introduction to Computer Programming □ Overview of programming languages. □ Programming and the Web: Introduction to web programming. 1.3.Hypertext and the World Wide Web □ Understanding hypertext and its significance. □ Introduction to the World Wide Web (WWW). □ Evolution of the WWW and its impact on society. 1.4.Introduction to HTML (Hypertext Markup Language) □ Basics of HTML syntax and structure □ Creating web pages with HTML □ Essential HTML tags and their functionalities 1.5.Programming vs. Scripting □ Understanding the difference between programming and scripting. □ Overview of scripting languages commonly used in web development. □ Role of scripting languages in web applications. 1.6.Client/Server Model. □ Overview of the client/server architecture. □ Understanding the roles of clients and servers in web applications.	CLO 1	Lecture, PBL, lab

❑ Interaction between clients and servers in the web development process. 1.7.Universal Addressing ❑ Introduction to the concept of universal addressing ❑ Overview of TCP/IP (Transmission Control Protocol/Internet Protocol) ❑ Understanding DNS (Domain Name System) and its role in mapping domain names to IP addresses 1.8.Universal Protocols ❑ Overview of HTTP (Hypertext Transfer Protocol). ❑ Understanding URLs (Uniform Resource Locators) and their structure. ❑ Introduction to HTML as a universal markup language. ❑ Overview of FTP (File Transfer Protocol) for file management on the web. 1.9. Website evaluation (readability, layout, coloring, trust...) 1.10. Basic UI/UX Design and concept ❑ Wireframe ❑ Prototype Laboratory contents (SEO, DNS, TCP/IP, URL, Wireframe and prototype)		
Chapter 2: Version control(5 hrs) 2.1 Introduction to version control systems and collaboration tools such as Git, GitHub, Trello Laboratory contents (create and use repository)	CLO6	Lecture, Laboratory, project

Chapter 3: WEB DESIGN AND DEVELOPMENT FUNDAMENTALS (12 hrs.) 3.1. Introduction to HTML. 3.2. Basic HTML Syntax and HTML Tags 3.3. HTML Document structure. 3.4. Text Formatting and structure. 3.5. HTML frames implementations, pros and cons 3.6. Working with Links and Images. 3.7. Lists and Tables. 3.8. Forms and input elements. 3.9. Multimedia and Embedded Content 3.10. SEO 3.11. HTML5 and Modern HTML Features. ✦ New semantic elements: <header>, <nav>, <section>, <footer>, etc. ✦ Multimedia elements: <video>, <audio>. ✦ Canvas for graphics and animations. ✦ Responsive design and media queries. 3.12. HTML best practices and Optimization Laboratory contents (write HTML code, HTML formatting tags, paragraph, list, heading, semantic tags, table, form, SEO, validation, develop website, version control systems)	CLO 2	Lecture, PBL, lab
Chapter 4: CASCADED STYLE SHEETS (CSS) (15 hrs.) 4.1. Introduction to CSS ❑ Why use style sheets? ❑ Advantages of CSS over inline styling and HTML styling. ❑ HTML vs. CSS: Separation of structure and presentation. 4.2. Anatomy of a CSS Rule ❑ Selectors: ID selectors, class selectors, element selectors. ❑ Declarations: Property and value pairs. ❑ CSS comments. 4.3. Inline CSS vs. Internal CSS vs. External CSS ❑ Inline styling within HTML tags ❑ Internal CSS using <style> tags in HTML. ❑ External CSS linking with <link> tag. 4.4. Style Sheet Rules ❑ Style Inheritance ❑ Style Rules Precedence	CLO 2	Lecture, PBL, Lab and Guest Lecture

4.5.CSS basics: Classification □ CSS Selectors and Combinators. □ Units of Measurement/CSS dimension (Absolute units, Relative units, Flexible units) □ Typography. □ Box model (Margins, Backgrounds, Text styling) □ Positioning and Layout Properties: Flexbox and Grid for modern layouts. □ Animation and Transition Properties. □ Responsive Design and media dependent cascades. □ Vendor Prefixes. 4.6.CSS Preprocessors □ SASS □ LESS 4.7.CSS standards of accessibility, usability and users 4.8. CSS Frameworks and Libraries □ Introduction to popular CSS frameworks like Bootstrap. □ Using pre-built styles and components for rapid development. Laboratory contents (Write CSS code, connecting CSS and HTML, CSS formatting, CSS layout, CSS animations, CSS transitions, CSS media dependent, CSS Preprocessors, using CSS frameworks)	CLO 2	Lecture, PBL, Lab and Guest Lecture
Chapter 5: Client-Side Programming -JavaScript (40 hrs.) 5.1. Introduction to JavaScript □ Overview of JavaScript and its role in web development □ Setting up the development environment □ Basic syntax and variables □ Data Types and Operators: Primitive data types (strings, numbers, Booleans), Complex data types (arrays, Objects), Operators and expressions □ Control Flow and Loops: Conditional statements (if, else if, switch), Looping structures (for, while, do-while, for-Each, for-Of, for-In), Iterating through arrays and objects □ Functions: Defining and calling functions, Parameters and return values, Function scope and closures, Higher-order functions and callbacks □ Arrays and Objects: Manipulating arrays (adding, removing, updating elements), Array methods (map, filter, reduce), Working with objects (creating, accessing properties)	CLO 2	Lecture, PBL, Lab, Guest Lecture

5.2.DOM Manipulation and Events ❑ Introduction to the Document Object Model (DOM) ❑ Accessing and modifying HTML elements ❑ Manipulating CSS classes and styles ❑ Event handling and event listeners 5.3. Asynchronous JavaScript ❑ Introduction to asynchronous programming ❑ Callback functions and the event loop ❑ Promises and async/await syntax ❑ Making API requests with Fetch 5.4. Error Handling and Debugging ❑ Common JavaScript errors ❑ Debugging techniques and tools ❑ Error handling with try...catch 5.5.ES6 and Modern JavaScript ❑ Introduction to ES6 features (let, const, arrow functions) ❑ Modules and import/export ❑ De-structuring assignments and spread syntax ❑ Classes and object-oriented programming (OOP) 5.6. Preserving data in browsers ❑ Cookies ❑ Local storage 5.7. Introduction to Front-End Development Tools ❑ Package managers (e.g., npm, Yarn) ❑ Bundlers (e.g., Webpack, Parcel) 5.8. JavaScript Best Practices and Code Organization ❑ Code formatting and linting. ❑ Modular code structure and organization. ❑ Code documentation and commenting. 5.9. Introduction to JavaScript Frameworks/Libraries ❑ Overview of popular JavaScript frameworks/libraries (React, Vue, Angular). ❑ Building a simple application using a framework/library. Laboratory contents (Implement JS variables, datatype, operators, conditionals, loops, array, string, function, OOP, form validation, DOM, API, cookies, Local storage, Bundler (SPA), Module import and export, De-structuring and structuring, Sync and Async, Error handling, debugging and test, using JS frameworks)	CLO 2	Lecture, PBL, Lab, Guest Lecture
--	---------------------	--

Transfer Skill: Problem analysis, problem-based learning (PBL) skills, algorithm analysis, Develop and implement web development.

IV. Assessment Method

No	Assessment type	Value/Mark	CLO
1	Assignment	10%	1
3	Mid-Exam	20%	1,2
4	Project	30%	1,2,3,4,5,6
5	Final Exam	40%	1,2,3,4,5,6

V. Text Book and references Text book

1. Michael Mendez, “The Missing Link, An Introduction to Web Development and programming” (2014).

References

1. Mozilla Developer Network: <https://developer.mozilla.org/enUS/docs/Learn/JavaScript>
2. <https://getbootstrap.com/docs/5.3/getting-started/introduction/>
3. <https://javascript.info/>
4. Holzner, ”HTML, Black Book”
5. Danny Goodman and Michael Morrison, “JavaScript Bible” 6th edition
6. <https://git-scm.com/>
7. <https://docs.github.com/en/get-started/quickstart/hello-world>